

LLM AGENTS • PLANNING • METACOGNITION

KnowSelf: Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness: autonomous regulation of knowledge utilization through situational self-awareness.

Source: arxiv.org/abs/2504.03553

PROBLEM FRAMING

Current agent training uses “flood irrigation” knowledge injection.

01

Trajectory imitation

Gold trajectories are injected as patterns to imitate, regardless of whether the agent needs help at a specific step.

02

Trial-and-error refinement

External feedback drives refinement, but the agent is not asked to judge whether reflection is sufficient.

03

Knowledge-augmented planning

Domain knowledge is applied broadly, increasing inference cost and sometimes hurting planning behavior.

Observed failure mode

- Fragile planning and pattern collapse on unexpected inputs.
- Unnecessarily high inference cost from indiscriminate support.

VS.

Human-like alternative

- Assess whether to act directly, pause and reflect, or seek external help.
- Regulate cognitive resources according to situational need.

THREE-SITUATION TAXONOMY

KnowSelf separates direct action, self-correction, and external knowledge need.

At each decision step:

Can the agent reach the gold action using its current policy and reflection?

Predicted action matches gold?

Yes → direct execution. No → test reflection.

Reflection fixes action?

Yes → self-correct. No → retrieve knowledge.

Fast thinking**FAST**

Directly produce the correct action with minimal deliberation.

Slow thinking**SLOW**

Initially incorrect, then correct after self-reflection.

Knowledgeable thinking**KNOW**

Still incorrect after reflection; consult external knowledge.

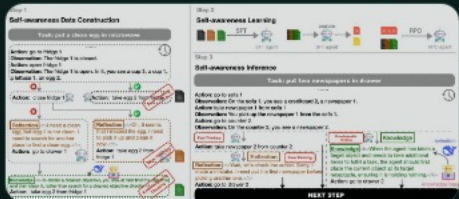
The decision criterion is operational: match gold action → fast; reflection fixes → slow; otherwise → knowledgeable.

FULL PIPELINE

KnowSelf: data construction → two-stage training → self-aware inference.

1. Self-awareness data construction

- Self-explore environments.
- Heuristic situation judgment at each step.
- Insert special tokens for fast, slow, or knowledgeable thinking.



2. Two-stage training

- SFT teaches initial self-awareness patterns.
- RPO refines generation of appropriate special tokens.

3. Self-aware inference

- Generate a situation token during deployment.
- Token triggers direct action, reflection, or knowledge retrieval.
- Knowledge is used selectively rather than by default.

QUALITATIVE GROUNDING

KnowSelf recovers task-relevant actions where strong reasoners fail.

Task A: "put a hot mug in cabinet"

OpenAI-01 After opening the microwave, incorrectly tries to take an egg from it.

KnowSelf Uses a Reflection token and realizes the mug should be heated using the open microwave.

Mode: slow thinking – reflection fixes the initially wrong action.

Task B: "put a saltshaker in drawer"

DeepSeek-R1 Gets stuck by reflecting that the open drawer should be closed first, which fails.

KnowSelf Applies a Knowledge token: place the object without removing unrelated items.

Mode: knowledgeable thinking – external knowledge resolves the situation.

Action: take egg 1 from microwave 1

Observation: Nothing happens.

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(b) Task: put a saltshaker in drawer

ALFWORLD RESULTS

KnowSelf reaches 84.33% with Llama-8B using only 15.01% knowledge.

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

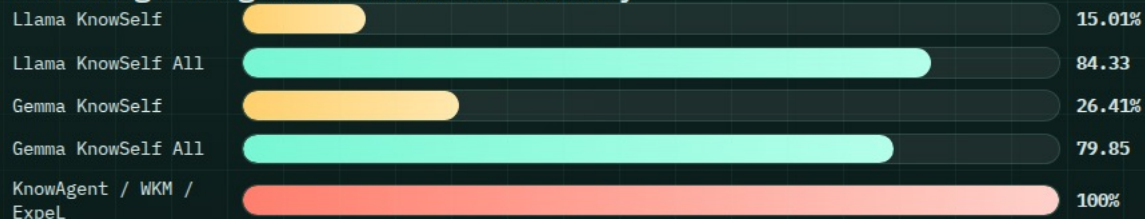
Success rates (%) across six ALFWORLD task types. "Know%" = proportion of steps using external knowledge.

Highlighted rows: KnowSelf.

EFFICIENCY AND SCALING

Selective knowledge beats blanket knowledge.

Knowledge usage vs. overall accuracy



The 100% row summarizes methods that use external knowledge at every step in the reported table.

84.33%

Llama-8B KnowSelf

Outperforms Llama-8B ETO (78.36), KnowAgent (75.37), WKM (77.61), and ExpeL (41.04) while using 15.01% knowledge.

79.85%

Gemma-2B KnowSelf

Matches GPT-4o ExpeL overall and exceeds all Gemma-2B baselines while using 26.41% knowledge.

Scaling behavior

The self-awareness mechanism remains effective across model sizes: a smaller Gemma-2B agent retains competitive performance through selective knowledge use.

Efficiency claim

KnowSelf's gains come from knowing when to retrieve, not from maximizing retrieval frequency.

TAKEAWAYS AND FUTURE DIRECTIONS

Self-awareness is the missing control layer in agent planning.

- 1 Indiscriminate knowledge injection is wasteful and can be suboptimal for interactive planning.
- 2 Agents can learn situational self-awareness with special tokens and lightweight heuristic labeling on self-explored trajectories.
- 3 Llama-8B KnowSelf surpasses GPT-4o ReAct and GPT-4o Expel: 84.33% vs. 76.12% and 79.85%.
- 4 Only 15.01-26.41% knowledge usage is needed versus 100% for knowledge-heavy competing methods.
- 5 The paradigm generalizes across model scales: Gemma-2B KnowSelf reaches 79.85%, matching GPT-4o Expel.

Future direction: treat reflection and retrieval as learned control decisions, not always-on add-ons.